

Detailed Diagnostic Report

PROPERTY DETAILS

- * Property Type: Flat
- * Property Age: Not Available
- * Number Of Floors: 11
- * Previous Repairs: No
- * Inspection Date: 27.09.2022

EXECUTIVE SUMMARY

The inspection and thermal reports indicate that the property is experiencing dampness, leakage, and cracking issues in various areas, including the hall, bedroom, kitchen, and common bathroom. The thermal analysis reveals temperature differences ranging from 5 to 5.4 degrees Celsius, indicating moderate to high severity issues. The overall condition of the property suggests a need for immediate attention to address these defects and prevent further deterioration.

AREA WISE FINDINGS TABLE

Area	Observation	Severity	Probable Cause
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Hall	Dampness at skirting level	Moderate	Concealed plumbing leakage
Bedroom	Dampness at skirting level	Moderate	Concealed plumbing leakage
Kitchen	Dampness at skirting level	Moderate	Concealed plumbing leakage
Common Bathroom	Tile hollowness and plumbing issue	Moderate	Poor tile jointing and concealed plumbing leakage
External Wall	Cracks and efflorescence	Moderate	Water seepage and poor maintenance
Parking Area	Seepage	Moderate	Poor waterproofing and drainage

OBSERVED DEFECTS

Dampness Related Issues

- * Observation: Dampness at skirting level in hall, bedroom, and kitchen
- * Severity: Moderate
- * Technical Impact: Potential damage to finishes, furnishings, and structural elements

Leakage Related Issues

- * Observation: Leakage in common bathroom and parking area
- * Severity: Moderate
- * Technical Impact: Potential damage to structural elements, finishes, and furnishings

Crack Related Issues

- * Observation: Cracks on external wall and bedroom wall
- * Severity: Moderate
- * Technical Impact: Potential weakening of structural elements and water seepage

Tile Joint Issues

- * Observation: Gaps between tile joints in common bathroom and master bedroom bathroom
- * Severity: Moderate
- * Technical Impact: Potential water seepage and damage to tile joints and surrounding areas

Surface Deterioration Issues

- * Observation: Efflorescence on external wall and dampness on common bathroom ceiling
- * Severity: Moderate
- * Technical Impact: Potential damage to surface finishes and structural elements

THERMAL ANALYSIS

The thermal data indicates temperature differences ranging from 5 to 5.4 degrees Celsius, with severity classifications ranging from low to high. The hotspots and coldspots are distributed across various areas, including the hall, bedroom, kitchen, and common bathroom.

- * Hotspot Temperature: 28.8 degrees Celsius
- * Coldspot Temperature: 23.4 degrees Celsius
- * Temperature Difference: 5.4 degrees Celsius
- * Severity Classification: High

The thermal evidence supports the inspection findings, indicating moderate to high severity issues related to dampness, leakage, and cracking.

ROOT CAUSE ANALYSIS

Issue: Dampness at skirting level

Probable Cause: Concealed plumbing leakage

Supporting Evidence: Inspection report and thermal analysis

Impact On Structure: Potential damage to finishes, furnishings, and structural elements

Issue: Leakage in common bathroom and parking area

Probable Cause: Poor tile jointing and concealed plumbing leakage

Supporting Evidence: Inspection report and thermal analysis

Impact On Structure: Potential damage to structural elements, finishes, and furnishings

Issue: Cracks on external wall and bedroom wall
Probable Cause: Water seepage and poor maintenance
Supporting Evidence: Inspection report and thermal analysis
Impact On Structure: Potential weakening of structural elements and water seepage

STRUCTURAL RISK ASSESSMENT

Issue	Risk Classification	Justification
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Dampness at skirting level	Moderate	Potential damage to finishes, furnishings, and structural elements
Leakage in common bathroom and parking area	Moderate	Potential damage to structural elements, finishes, and furnishings
Cracks on external wall and bedroom wall	Moderate	Potential weakening of structural elements and water seepage

RECOMMENDED REPAIR METHODOLOGY

Repair Procedure: Waterproofing and tile jointing repairs
Materials Required: Waterproofing membranes, tile adhesives, and grouts
Priority Level: Immediate

Repair Procedure: Plumbing repairs and replacement
Materials Required: Plumbing fixtures and piping
Priority Level: High

Repair Procedure: Crack repair and surface finishing
Materials Required: Crack sealants, surface finishes, and paints
Priority Level: Medium

RISKS DUE TO DELAY

Delayed repairs may lead to further deterioration of the property, resulting in increased repair costs, potential damage to structural elements, and health risks associated with dampness and water seepage.

COST IMPACT ASSESSMENT

The cost impact of delayed repairs is expected to be medium to high, with potential costs increasing exponentially if left unaddressed. The costs may include repairs to structural elements, replacement of finishes and furnishings, and potential health risks.

FINAL CONCLUSION

Based on the inspection and thermal reports, it is recommended that immediate

attention be given to addressing the dampness, leakage, and cracking issues in the property. A comprehensive repair methodology, including waterproofing, tile jointing, plumbing repairs, and crack repair, should be implemented to prevent further deterioration and potential damage to structural elements. Delayed repairs may result in increased costs and potential health risks, emphasizing the need for prompt action.